Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Method of separation of variables

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Method of separation of variables

1 Method of separation of variables

Study focus: Method of separation of variables. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000074 · EV0000075

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In last two classes, we learned how to solve problems on Lagrange's linear equation by using method of multipliers or method of grouping or both.

Key details

- In some special cases, we also applied differential method.
- So in this session, we will learn what is a method of separation of variables.
- How do we apply this for a problem?
- And what is its advantage.
- As an application,.
- Separation of variables provides closed form solutions for many fundamental partial differential equations, namely one dimensional heat equation, the wave equation, and the Laplace equation.
- By reducing each PDE to a set of ordinary differential equations.
- In a future class, we will see how to apply this method of separation of variables to find out the various possible solutions of one dimensional heat equation.
- Before that, we'll understand.
- How to apply this method for a problem.
- U equals f of x y as usual u is dependent and x and y are independent variables.
- Then the partial derivatives present in the equation should be.
- reduced to ordinary derivatives.
- Assuming u is a function of x and y.
- ordinary partial derivative into ordinary derivative in a similar manner if we have dou u by dou y.

- Because y is x y, this can be written as in this case we take x outside and written as d capital y divided by d small y.
- In terms of ordinary derivative.
- Or the partial derivatives are reduced to ordinary derivative in this manner.
- In other words.
- The expression containing ordinary derivatives should be rearranged in such a way that left hand side is a function of x only and rhs is a function of y only.
- That's what you can see in expression number two.

Example or application from the lecture

- This method is also called product method because we assume the solution of the given PDE in the form of a product is depending on only one independent variable.
- Let us assume that.
- If the given PD contains x and y to be independent variables, we need to assume the solution of the given PDE in the form of a product that is u equals x of x times y of y where x is a function of x only and this is a function of y only.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 05:00–09:59

After we have this scenario that is f of x equals g of y, so mathematically they both have some common constant k.

Key details

- Therefore, we equate both sides to some common constant k.
- Remember, this is nothing but ordinary derivatives.
- We need to solve these two by known methods.
- We solve these two for capital X and capital Y.
- It means the one what you got from step number two.
- So this is how we apply method of grouping, sorry, method of separation of variables for a problem.
- Use the method of separation of variables to solve this partial differential equation, which is nothing but three times dou u by doux plus two times.
- dou u by dou y equals zero so conditions are u of x comma zero equals four of e power minus x so it is not necessary to have conditions like this we can apply this method for problems without this condition also.
- That is u equal to x of x times y of y.
- So it means as we know from the procedure.
- This three dou by dou x becomes three capital y e capital X divided by D small X plus second term will be two times X into P y by.
- P y equals zero.
- This is what we assume.
- Then dx by x.

- We just ship this right hand side.
- Then we get dy by y.
- It means we write one by x by t x equals k.
- Another partition minus two by three.
- One by capital Y by DY equals k.

Example or application from the lecture

- Let us say that f of x equals k as well as g of y equals k.
- In order to understand this procedure, let's look at one or two examples.
- So let's see how do we use this condition in later stage?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 10:00–14:59

K1 constant for convenient purpose.

Key details

- Then X becomes C1 E power KX where C1 becomes e power K1.
- So this is the first solution.
- After integration it is log y equals minus 3 by 2 k y plus some constant k2.
- Therefore y equals c2 times e power minus 3 by 2 times y.
- U equals XY.
- E power A times E power B minus B, E power A minus B.
- You can take K.
- outside so that is same as x minus three by two of y so in the problem if you don't have any conditions like this.
- we can say we can stop here.
- So whenever we have a condition we need to proceed.
- Obviously, we need to find out a value as well as this k value.
- So when we apply the condition, if you observe here 4e power minus 6 equal to u of x comma 0, it means wherever we have y, we put 0.
- I mean if you substitute both A and C K value.
- We get this.
- I think now you understood the meaning of this statement in the pro in the data.
- It means by reducing each PDE to a set of ordinary differential equations as a each depending on only one independent variable.
- So it means u of x dot x dot y.
- And this is how we can convert partial derivative into ordinary derivative.
- And the condition is u of x.
- comma zero equals three times e power two x.
- So to eliminate y in the first and x in the second, we should divide through by the assumption, which is xy.

Example or application from the lecture

- That is u of x y to be the this is a solution for the given problem.
- Seven is substitute both in the given problem.
- So finally this is a solution for the given problem by using method of separation of variables.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 15:00–19:59

four by y by.

Key details

- It is not compulsory.
- Therefore x is some a1 k.
- So this is the first solution.
- or this is a solution of this ordinary differential equation that is 1 by x dx by x.
- In a similar manner how do we solve second one?
- Software solving for Y.
- So when we use a condition.
- We are equating this part to three times e power two x equals c as it is.
- Of course in our case, okay, you can call some c e power kx.
- When y is zero, so this part disappears.
- it becomes u equals three e power this is actually two of two of x plus pi by four times y.
- So therefore three power this is two x plus five by two y.
- So in this session we understood what is the method of separation of variables, what happens, and we also learned how to solve problems using this method.
- So in the next class, we apply this method to find out the various possible solutions of one dimensional heat equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Method of separation of variables transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...

C Revision and retrieval practice